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Exploring Atlantic Sturgeon Migration and Habitat in the Roanoke River

Introduction:

Atlantic sturgeon (*Acipenser oxyrinchus oxyrinchus*) are a large, long-living, species of fish found along the East Coast of the United States and Canada. They are anadromous, meaning that they spawn in freshwater before migrating to grow in brackish and salt water environments. They can grow as long as 14 feet, weigh up to 800 pounds, and live as old as 60 years (NOAA Fisheries). They are currently found to migrate in 22 of the 38 historical spawning rivers throughout their range, which is split into five distinct population segments, or DPS. The species is classified as endangered in the the New York Bight, Chesapeake Bay, Carolina, and South Atlantic DPS, and threatened in the Maine DPS.

The sturgeon travel to the upper reaches of the rivers to spawn, at or near the Atlantic seaboard fall line. Hard substrate is required so the eggs have a stable surface to adhere to. This type of substrate is also vital during the larval stage, as they can hide in the spaces between rocks to avoid predation during migration downstream. Freshwater is required during the first few weeks of life, and by age two they become more salt tolerant and are able to enter marine waters.

Water depth and water quality are important physical features to allow clear passage through rivers for both adults migrating to spawning sites and juveniles moving back to sea. Water depth must remain at least 1.2 meters deep at all times to sustain sturgeon at any life stage, and low turbidity is important for increasing egg survival and success in foraging for food.

Fishing activity and dams along the river pose two major difficulties for sturgeon migration, where the dams physically impede access to spawning and foraging, and fishing can cause unintentional hooking, entanglement, and collisions with vessels. Other threats include dredging, beach nourishment projects, in-water construction, and outfalls/discharge (NOAA Fisheries).

Taking these physical requirements and obstacles into consideration, the following research intends to analyze the way these factors impact the upstream migration of these fish to spawning sites by creating a least cost path map showing which areas of rivers are more suitable for spawning/movement, and which parts pose the greatest threat to their livelihood. The Roanoke River in North Carolina, part of the Carolina DPS, is the area of interest in this study as it is widely

considered to contain the most Atlantic sturgeon in the state with a known, active spawning population. Substrate type, water turbidity, proximity to dams, proximity to public fishing areas, and water depth will be used to calculate the suitability of the river for migration. Most of the analysis is completed in ArcGIS Pro version 3.5, with analysis of satellite imagery completed in Google Earth Engine.

Study Site:

The Roanoke River basin begins in the Blue Ridge Mountains in Virginia, crosses into North Carolina, and empties into the Albemarle Sound. The total length is 410 miles long, with approximately 130 of those in North Carolina. The study area covers the portion of the river running from the fall line to the Albemarle Sound, which flows through five North Carolina counties: Halifax, Northampton, Bertie, Martin, and Washington (*Roanoke River Basin Documents*). The study site is further narrowed down to an area of 250 meters on either side, leaving a 500 meter wide corridor in total to lower processing times.

Data:

County DEMs

Five digital elevation models for the five counties in the study site were downloaded from the North Carolina Spatial Data download. The resolution was 3 foot pixel sizes.

Fall Line

Accessed from ArcGIS Online and downloaded in vector polyline format. Showed approximate location of the Atlantic seaboard fall line of the entire East Coast of the U.S.

Fishing Area Locations

Vector point data of public fishing locations in North Carolina. Projection is in Web Mercator. Accessed from the NC Portal Access to Wildlife Systems.

Dam Locations

Vector point data of dam locations in North Carolina rivers. Projection is in Web Mercator and data was downloaded from the NCDEQ GIS Open Data Portal.

Remote Sensing Data

Copernicus/Sentinel 2 MSI data accessed through Google Earth Engine and downloaded to be used in ArcGIS Pro after being processed.

State Hydrography

Vector polygon feature layer downloaded from NC OneMap. Shows major hydrography features of North Carolina, including streams and rivers. NAD83 State Plane projection.

Rivers

Downloaded from the New York State Department of State's Geographic Information Gateway. Data originally sourced from the National Marine Fisheries Service. Depicts the rivers part of the Carolina DPA of Atlantic Sturgeon, consisting of eight rivers across North and South Carolina. Vector line feature layer.

Methods:

The following steps are completed in ArcGIS. The Batch Project tool to reproject all input layers to EPSG:6543, or NAD83(2011)/North Carolina (US Feet). Pairwise Buffer is used to create a 250 meter buffer around the Roanoke River line to use as a mask in future processing. The 500m wide area is large enough to accommodate the widest part of the river toward the mouth at the Albemarle Sound. Extract by Mask is used on the five county DEMs with the 250m buffer as the mask so the processing time can be significantly shortened in the future. Select Layer by Location and Export Features is used to select the dams and fishing areas that fall within or on the line of the 250m buffer so that a new layer with only the relevant points is created. The five masked DEMs are combined using Mosaic to New Raster to create a single DEM layer (*Figure 1*). The percent rise of slope of the new DEM is calculated using the Slope tool (*Figure 2*).

Two points are created at the mouth of the Roanoke River into the Albemarle sound and again at the top of the river near the fall line and made into their own layer. Vertices are then added to the Roanoke River line every 100 meters using the Densify tool, before using the Split Line at Vertices tool on that output. Add Surface Information is used to add the Z_MIN, Z_MAX, and Z_MEAN fields to the new line segments, using the DEM as the input surface. Calculate Field is used to add the estimated water surface elevation, or WSE, using “!Z_MEAN! + 6.56” to add 2 meters (6.56 feet) to the riverbed elevation. This number was chosen due to the fish's need to have 1.2 meters depth to live, and the 2m gives wiggle room for fluctuating water levels. Generate Points Along Lines is used to create points every 100m along this segmented line (*Figure 3*). The IDW interpolation tool is used to continue working toward the river depth calculation (*Figure 4*), and Raster Calculator is used to approximate the water level (*Figure 5*), and create a new binary mask in which all pixels greater than or equal to 1.2m (3.94ft) are set to a value of 1, and all else 0.

The 250m river buffer is exported using Export Features and uploaded to a new Google Earth Engine project. The following section is completed using GEE. Sentinel-2 surface reflectance data is imported and dates are sorted to only include dates through October and November of 2022, or the two months with the least precipitation, and therefore the least clouds. Cloudy pixel percentage was also set to filter any with values higher than 20. The QA60 band was used to create a function to remove cloud and shadow interference, which was then applied to the image collection. Normalized difference water index (NDWI) is found by computing the normalized difference between band 3 and band 8, and this is used to create a mask that filters out all pixels that are not found to be water. Normalized turbidity difference index is found by calculating the normalized difference between bands 4 and 3. The turbidity values are masked to the water pixels only, and the visualizations parameters are set before the final result is added to the map frame (*Figure 6*). The data is exported as a layer to Google Drive, and then downloaded and imported back into the ArcGIS Pro project for further processing (*Figure 7*).

The following steps are completed back in ArcGIS Pro. Substrate type is estimated by using Raster Calculator on the slope layer, assuming slope lower than or equal to 0.2 is soft, or 1, lower than or equal to 0.5 is mixed, or 2, and everything else is hard, or 3. This output is clipped to the area of interest by using Extract by Mask (*Figure 8*). Raster Calculator is used to calculate min-max normalization on the new clipped turbidity raster to rescale its maximum and minimum values to equal 0 and 1 instead. The turbidity cost raster is then calculated using the equation “ $\text{Int}(1 + 9 * \text{“turbidity_norm”})$ ”.

The cost rasters for proximity to fishing areas and dams is created by first using the Distance Accumulation tool on each layer. Each layer is clipped to the AOI, and had their cost layers calculated. The fish cost (*Figure 9*) used “ $\text{Int}(1 + 5 * (1 - (\text{“fish_distance_clipped”} / 3281)))$ ”, and dam cost (*Figure 10*) used “ $\text{Int}(1 + 9 * (1 - (\text{“dam_dist_clipped3”} / 16404)))$ ”.

The Split editing tool is used to remove the half of the AOI polygon that is to the left of the Atlantic seaboard fall line, which will then be used to clip the final product. Raster Calculator is used to produce the friction layer using “ $(\text{“sub_cost”} + \text{“turb_cost”} + \text{“fish_cost”} + \text{“dam_cost”}) * \text{“depth_mask”}$ ” (*Figure 11*). Distance Accumulation is used to create the inputs (*Figure 12*) for the Optimal Path as Raster tool, which creates the final least cost path line of the river migration (*Figure 13*).

Results:



Figure 1: the clipped DEMs mosaiced together



Figure 2: calculated slope of the DEM



Figure 3: Points with WSE data generated along the river polyline



Figure 4: very zoomed in view of the output of the IDW tool



Figure 5: estimated depth raster



Figure 6: turbidity raster in Google Earth Engine

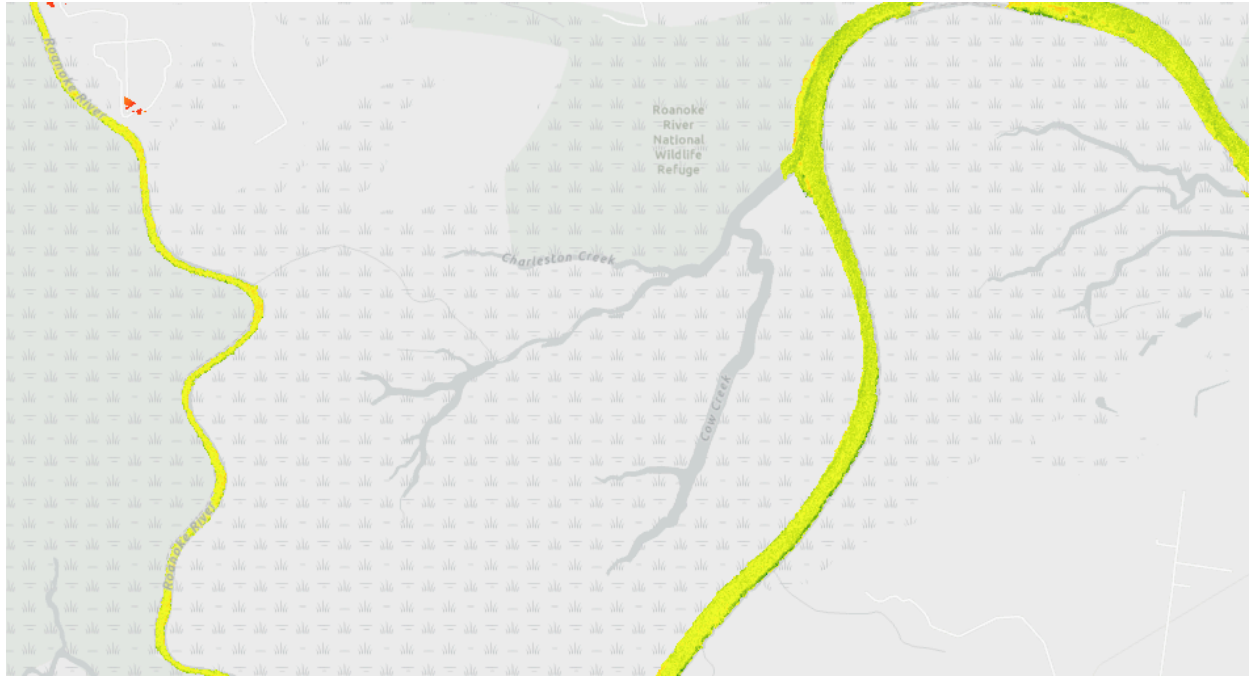


Figure 7: turbidity raster uploaded to ArcGIS Pro

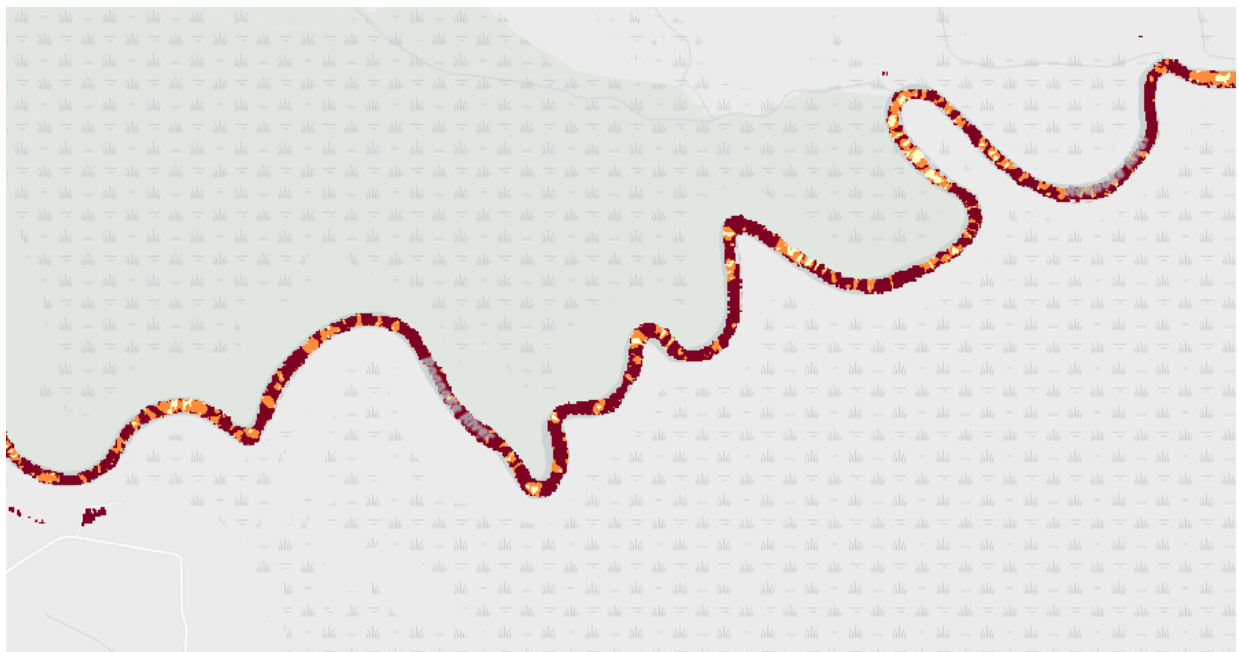


Figure 8: substrate cost layer

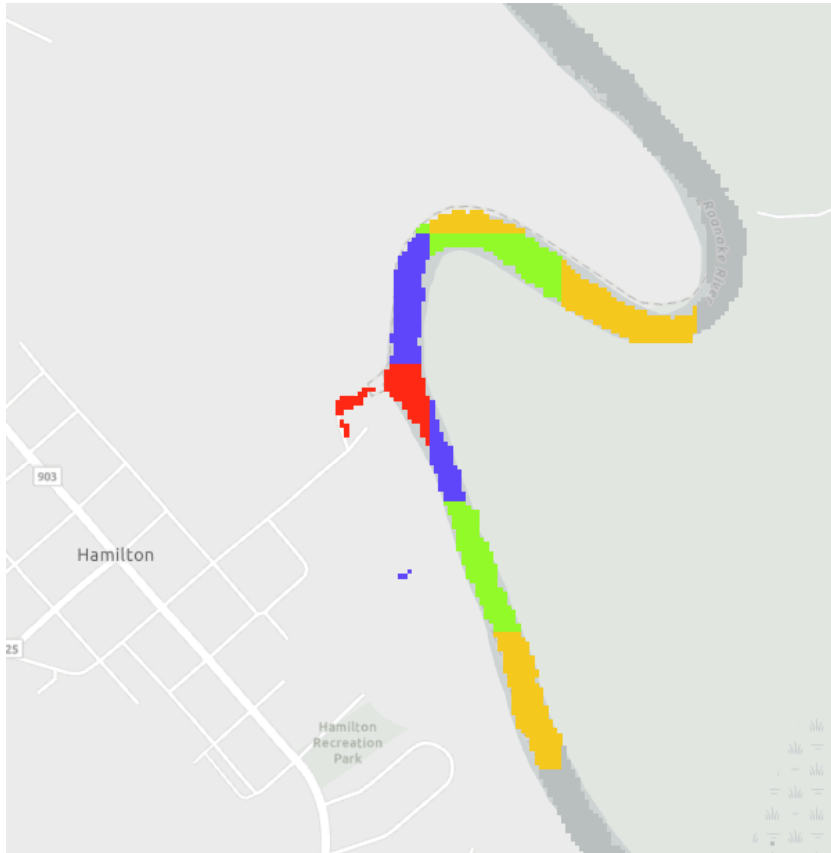


Figure 9: fishing area proximity cost layer

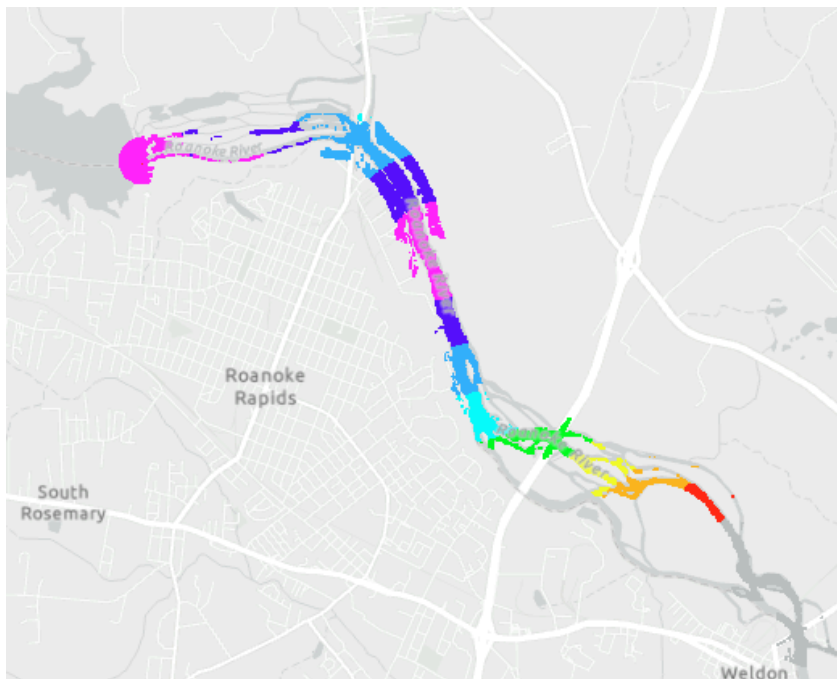


Figure 10: dam proximity cost layer

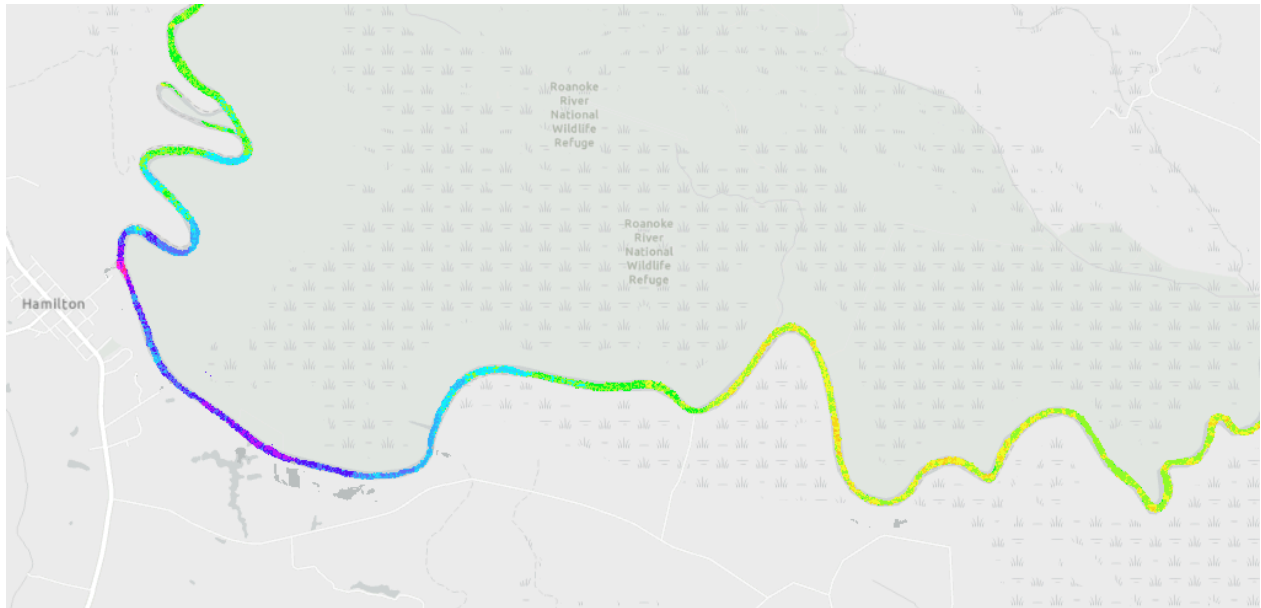


Figure 11: friction layer

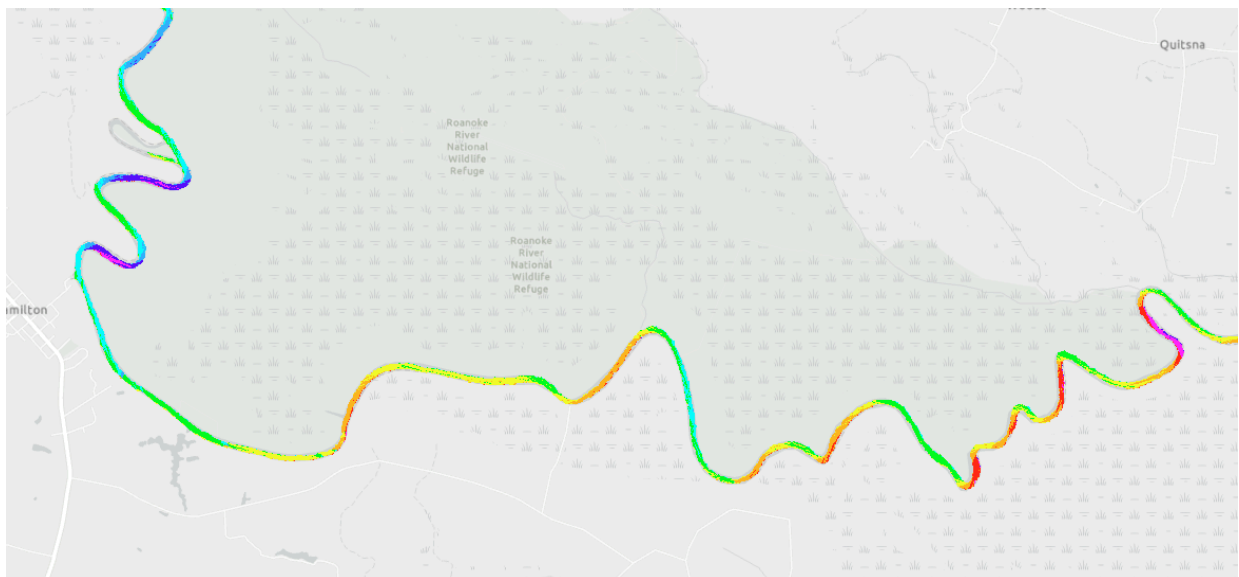


Figure 12: distance accumulation layer



Figure 13: final LCP line shown with destination point and friction layer

Link to AGOL page to download Map Package to view final product in detail:

<https://ncsu.maps.arcgis.com/home/item.html?id=47c8eb1abb0c41ea9f968d5654dee541>

The final product shows a significant choke point near the fall line in which there were no viable pixels of the waterway that were navigable for the fish, causing the initial location of the end point to move closer than where it was originally defined, right on top of the intersection of the river and the fall line. *Figure 13* shows this choke point and where the new end point was set to complete analysis, as the tool would not work if the pixels did not connect.

As expected, there is a great impact of water navigability/site suitability nearing dam locations, and slightly lesser so in proximity to fishing sites. These are the most notable barriers, but suitability does have more subtle overall changes when looking at larger sections of the river as a whole.

Discussion:

I was unable to find studies focusing on an area this specific and mapping these details alone and how they interact, so I am unable to form many comparisons on the exact nature of this map. I originally wanted to upload all the final layers to an AGOL map for the reader to be able to view easily, but I hit the limit trying to

upload the raster layers I wanted to show. The downloadable map package was the closest way I could let others view the results of the analysis.

The biggest problem I have here is that the site suitability itself is far more useful than the least cost path line itself, as the river is narrow in many places and the pixel size can only be so small. If I were to replicate these results again in the future, I would focus on a larger, wide-open area such as the Albemarle Sound or another open body of water that these fish have to navigate to find areas in which they are more likely to thrive. The ultimate goal was to identify areas in greater need of possible checkpoints and intervention during their migration and spawning season, as occasionally scientists will step in to aid their travels as their numbers are already threatened.

Conclusion:

The techniques used here produce decent results to identify trouble areas of the migration of Atlantic sturgeon upstream the Roanoke River in North Carolina, as well as provide a framework to apply these steps to another area of interest in the future. I expect to be able to reproduce similar results to these again in future work of mine.

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Acknowledgements:

Microsoft Copilot was used to troubleshoot ArcGIS Pro and Google Earth Engine code during the analysis phase.

Appendix:

Link to GEE code:

<https://code.earthengine.google.com/8b2ac9a051f54f5492a98fb079b22c5d>